Project Overview – GROUP 11

This report details the individual contributions of each team member to the **Toxicity Mining NLP** project, a study in categorising hate speech and toxic language across social media using the Google Jigsaw Civil Comments dataset (~1.8M records) and TweetEval benchmark (9K records). The final modelling dataset of **58,954 records** (51.2% non-toxic / 48.8% toxic) was used to train and evaluate **7 models across 4 ML categories**, achieving a best accuracy of **87.12%** and best F1 of **87.39%** with a fine-tuned DistilBERT transformer.

Individual Contributions

Himanshu Jain

- Conducted full Exploratory Data Analysis across Jigsaw (~1.8M) and TweetEval (9K) datasets
- Identified cross-platform domain gap: only 2.6% vocabulary overlap between forum text and tweets
- Performed class-distribution analysis confirming 51.2% / 48.8% non-toxic / toxic label split
- Built KMeans clustering pipeline (k=2) with Elbow Method, Silhouette Score, and Davies-Bouldin validation
- Applied t-SNE dimensionality reduction to produce 2D cluster visualisations confirming non-linear separation
- Conducted vocabulary-frequency analysis revealing distinct toxicity lexicons across domains
- Detected and removed text outliers using IQR-based method; cluster split (93.5% / 6.5%) mirrored class distribution

Key Notebooks / Deliverables

- clustering.ipynb (KMeans, t-SNE)
- EDA notebooks, class distribution & vocabulary analysis
- Outlier detection (IQR method)
- Presentation slides (all milestones)

Key Results

- KMeans k=2, Silhouette Score = 0.0999
- Cluster split: 93.5% / 6.5%
- 2.6% cross-platform vocab overlap

Rahul Hipparkar

- Fine-tuned DistilBERT transformer on the toxic comment dataset achieving the best accuracy (87.12%) and F1 (87.39%)
- Implemented Naive Bayes classifier (77.91% accuracy, 77.91% ROC-AUC) as the probabilistic baseline
- Built LightGBM gradient-boosting model (76.99% accuracy) with Optuna hyperparameter optimisation (50 trials)
- Designed Stacking Classifier combining Naive Bayes, LightGBM, and Logistic Regression as meta-learner (82.29% accuracy)
- Discovered 494 toxic vs 454 non-toxic frequent itemsets; top rule: {black} -> {white} with lift = 6.51

Key Notebooks / Deliverables

- ToxicityMining_NaiveBayes.ipynb
- ToxicityMining_LightGBM.ipynb
- ToxicityMining_Stacking_Classifier.ipynb
- ToxicityMining_DistilBERT.ipynb
- Model_Comparision.ipynb (metrics aggregation)

Key Results

- TF-IDF: 50K features, unigrams + bigrams
- SVD: 50,000 → 200-300 dims
- GridSearchCV applied to NB & LR
- DistilBERT: 87.12% acc, 87.39% F1
- Stacking: 82.29% acc, ROC-AUC 0.898

- Confirmed DistilBERT as the recommended production model, outperforming all classical approaches by 5+ accuracy points
- Set up DistilBERT WordPiece tokenisation pipeline (max 128 tokens) for transformer input
- Conducted hyperparameter tuning via GridSearchCV for Naive Bayes and Logistic Regression models

Augustine Antony Joy

- Implemented Frequent Pattern Mining using Apriori algorithm (min_support=0.01, min_confidence=0.2)
- Built TF-IDF vectoriser with 50,000 features (unigrams + bigrams) for classical ML models
- Applied Truncated SVD to reduce feature space from 50,000 to 200-300 dense dimensions
- Configured StandardScaler normalisation to standardise SVD-transformed features
- Designed the end-to-end NLP preprocessing pipeline: contraction expansion, lowercasing, mention/URL normalization
- Implemented NLTK tokenisation and stopword removal across the full 58,954-record modelling dataset

Key Notebooks / Deliverables

- ToxicityMining_FrequentPatternMining.ipynb
- Data Scrapping

Key Results

- Apriori: 494 toxic itemsets, lift = 6.51

Glen Vadakkoott

- Implemented Logistic Regression classifier (83.90% accuracy, 91.49% ROC-AUC) as standalone model and Stacking meta-learner
- Led Frequent Pattern Mining coordination and Apriori result analysis across toxic vs. non-toxic discourse
- Demonstrated that toxic discourse exhibits denser political/racial co-occurrence patterns than non-toxic text
- Maintained GitHub repository, coordinated PR reviews (#4 clustering, #5 pattern mining), and enforced coding conventions
- Authored Milestone reports and maintained README, project status, and milestone-tracking documentation
- Built and maintained the project website (glenpaulson.github.io/toxicity-mining-website)
- Created and updated all presentation slides for Milestone demos and the final poster

Key Notebooks / Deliverables

- ToxicityMining_LogisticRegression.ipynb
- Milestone_2 data-prep & EDA notebooks
- Project website & GitHub repo management

Key Results

- LR: 83.90% acc, 91.49% ROC-AUC (best ROC)
- GitHub: repo init, PRs #4 & #5 merged
- Website: glenpaulson.github.io/toxicity-mining-website

Model Implementation & Ownership

Model	Category	Owner	Key Result / Notes
Naive Bayes	Classification	Rahul Hipparkar	77.91% acc | GridSearchCV tuned
LightGBM	Classification	Rahul Hipparkar	76.99% acc | Optuna 50 trials
Stacking Classifier	Ensemble	Rahul Hipparkar	82.29% acc | NB + LGBM + LR meta
DistilBERT	Deep Learning	Rahul Hipparkar	87.12% acc | 87.39% F1 (best overall)
TF-IDF + SVD Pipeline	Feature Eng.	Rahul Hipparkar	50K features -> 200-300 dims (SVD)
Logistic Regression	Classification	Glen Vadakkoott	83.90% acc | 91.49% ROC-AUC (best)
KMeans Clustering	Unsupervised	Himanshu Jain	k=2 | Silhouette = 0.0999
Apriori / FP Mining	Frequent Patterns	Augustine Antony Joy	494 toxic itemsets | lift = 6.51

Project Milestone Progress

Milestone	Description	Status
Milestone 0	Project Proposal & Team Formation	**Completed**
Milestone 1	Project Framing & Website Launch	**Completed**
Milestone 2	Data Preparation, Collection & Cleaning	**Completed**
Milestone 3	Model Implementation & Evaluation	**Completed**
Milestone 4	Final Report & Conclusions	**Completed**